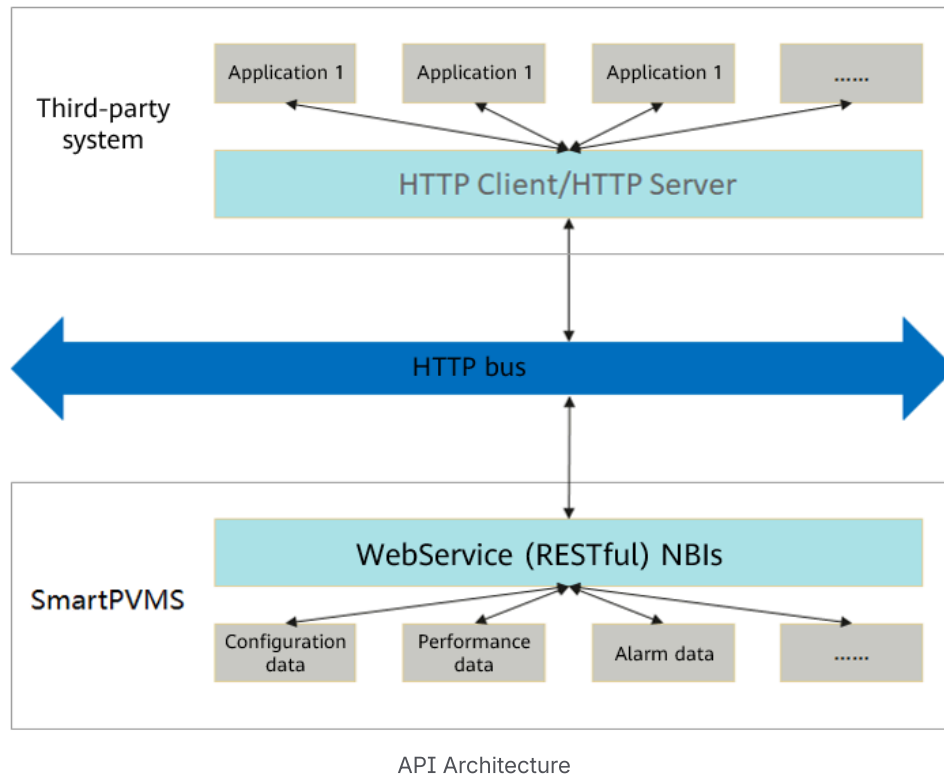


Integration Guide: Huawei SmartLogger via SmartPVMS Northbound REST API

This document outlines how to connect and communicate with Huawei SmartLogger devices through the SmartPVMS (≥ v24.4.0) northbound REST API for reading device data and issuing control commands (e.g. active power limit).



1. Prerequisites

- SmartLogger3000 or equivalent devices must be registered in a SmartPVMS or FusionSolar platform (version 24.4.0 or higher).
- API access must be enabled with credentials obtained through Huawei's FusionSolar/SmartPVMS web portal.
- Scopes required:
 - `pvms.openapi.basic` for data queries
 - `pvms.openapi.control` for control operations

2. Authentication

2.1 OAuth2 Flow (Recommended)

1. Redirect the user to the authorization URL:

```
1 GET https://oauth2.fusionsolar.huawei.com/rest/dp/uidm/oauth2/v1/authorize
2   ?response_type=code
3   &client_id={CLIENT_ID}
4   &redirect_uri={REDIRECT_URI}
5   &scope=pvms.openapi.basic pvms.openapi.control
```

2. After user login, the browser will return:

```
1 https://{redirect_uri}?code={AUTH_CODE}
```

3. Exchange the authorization code for an access token:

```
1 POST https://oauth2.fusionsolar.huawei.com/rest/dp/uidm/oauth2/v1/token
2 Content-Type: application/x-www-form-urlencoded
3
4 grant_type=authorization_code
5 &code={AUTH_CODE}
6 &client_id={CLIENT_ID}
7 &client_secret={CLIENT_SECRET}
8 &redirect_uri={REDIRECT_URI}
```

4. Store the `access_token` and `refresh_token` . The access token is valid for 1 hour.

2.2 API User (Alternative)

- Log in via `/login` endpoint using API credentials created from the SmartPVMS admin panel.
- Use the session token returned in each API request header.

All requests must include:

```
1 Authorization: Bearer {access_token}
2 Content-Type: application/json
```

3. Reading Data

3.1 List All Plants

```
1 POST https://{PVMS_DOMAIN}/thirdData/stations
2
3 Body:
4 {
5   "pageNo": 1,
6   "gridConnectedStartTime": 0,
7   "gridConnectedEndTime": 999999999999
8 }
```

3.2 List Devices in a Plant

```
1 POST https://{PVMS_DOMAIN}/thirdData/devices
2
3 Body:
4 {
5   "plantCode": "{plantCode}",
6   "pageNo": 1
7 }
8
```

3.3 Query Device Data (e.g., inverter generation)

```
1 POST https://{PVMS_DOMAIN}/thirdData/deviceRealKpi
2
3 Body:
4 {
5   "deviceCode": "{deviceCode}",
6   "kpis": ["active_power", "reactive_power"],
7   "collectTime": "{timestamp}"
8 }
9
```

4. Control Commands

4.1 Set Active Power Limit

```
1 POST https://{PVMS_DOMAIN}/thirdData/controlDeviceActivePower
2
3 Body:
4 {
5   "plantCode": "{plantCode}",
6   "deviceCode": "{deviceCode}",
7   "mode": "LIMIT",
8   "activePower": 5000
9 }
```

4.2 Check Execution Status of a Task

```
1 POST https://{PVMS_DOMAIN}/thirdData/taskQuery
2
3 Body:
4 {
5   "taskId": "{taskId}"
6 }
```

5. Error Handling

Error Code	Meaning	Resolution
305	Token expired or invalid	Refresh token or reauthenticate
401	Permission denied	Verify scopes and account privileges
429	Too many requests	Apply backoff, throttle retries
500+	Server/internal error	Contact Huawei support if persistent

6. Example: cURL Commands

Get Token:

```
1 curl -X POST https://oauth2.fusionsolar.huawei.com/rest/dp/uidm/oauth2/v1/token \  
2 -H "Content-Type: application/x-www-form-urlencoded" \  
3 -d "grant_type=authorization_code&code={AUTH_CODE}&client_id={CLIENT_ID}&client_secret=  
  {CLIENT_SECRET}&redirect_uri={REDIRECT_URI}"
```

List Plants:

```
1 curl -X POST https://{PVMS_DOMAIN}/thirdData/stations \  
2 -H "Authorization: Bearer {access_token}" \  
3 -H "Content-Type: application/json" \  
4 -d '{"pageNo":1,"gridConnectedStartTime":0,"gridConnectedEndTime":999999999999}'  
5
```

Send Active Power Limit Command:

```
1 curl -X POST https://{PVMS_DOMAIN}/thirdData/controlDeviceActivePower \  
2 -H "Authorization: Bearer {access_token}" \  
3 -H "Content-Type: application/json" \  
4 -d '{"plantCode":"NE=12345678","deviceCode":"DL=87654321","mode":"LIMIT","activePower":5000}'  
5
```

7. Additional References

- Full API Reference Index:
<https://support.huawei.com/enterprise/en/doc/EDOC1100366278>
- API Account Management:
<https://support.huawei.com/enterprise/en/doc/EDOC1100366278/b71c4d05/flow-control-using-the-api-account>
- Device Control Overview:
<https://support.huawei.com/enterprise/en/doc/EDOC1100366278/93b3b44a/controlling-device-active-power>
- Task Result Queries:
<https://support.huawei.com/enterprise/en/doc/EDOC1100366278/3517f31f/querying-the-control-task-result>

